

Reachability Search over Convex Transition Sets for Multi-Regrasp Manipulation Planning

Julia López Gómez and Steve Tonneau

School of Informatics, University of Edinburgh, United Kingdom

[TODO: emails]

Abstract—Turning a cap, a valve or a knob further than the wrist can rotate requires the robot to grasp, turn, release and grasp again, as many times as the goal, the joint limits and the surrounding obstacles demand. Planning these multi-regrasp tasks is hard for existing methods. Sampling-based manipulation planners are only probabilistically complete, so they cannot conclude that a goal needs more grasps or is unreachable. Optimisation-based and mixed-integer planners must fix the number of grasps in advance and, on high-dimensional arms, rely on local solvers or on numerically fragile big-M formulations. We represent the manipulation graph with convex sets: a convex decomposition of the free space of robot and object, intersected with a convex set of grasping configurations, gives *transition sets* in which the robot can grasp, turn and release the object. We search them with an A* whose nodes pair a convex set with the interval of object configurations reachable inside it. When one object coordinate is coupled to one robot joint, this interval is propagated and compared exactly. The graph is built with linear programs only, and the search returns a plan with the minimum number of grasps relative to the sets, or proves that none exists, without fixing the number of grasps in advance. On a 10-DOF cap-turning task with a Franka Panda, where our earlier mixed-integer formulation is numerically unreliable, it plans in 4ms. Because probabilistic decompositions can contain collisions concentrated on the grasp sets, we validate the transition sets before searching them. [TODO: headline comparison with the MIQP and the sampling-based planner (E1–E4), and the validated Q0 result (E5).]

Index Terms—Manipulation planning, regrasping, reachability, graph search, convex decompositions.

I. INTRODUCTION

Many manipulation tasks require the robot to grasp the same object several times. To turn a bottle cap, a valve or a knob further than its wrist can rotate, a robot must grasp the object, turn it, release it, rotate its wrist back and grasp it again. How many grasps are needed is not known in advance. It depends on the goal, on the joint limits of the robot, and on obstacles near the object, which shorten the part of the wrist range that can be used without collision. A planner must decide how many grasps to make, where each one starts and ends, and how the robot moves between them without collision. Ideally, it should also be able to tell when a goal cannot be reached with fewer grasps, or not at all.

The task looks simple, but it combines three difficulties. *Grasps happen on thin sets*. For a 7-DOF arm holding a

cap, four equality constraints on the pose of the hand leave a curved, lower-dimensional set of grasping configurations in the 10-dimensional space of robot and object. The robot can only use the part of this set that is collision-free, which lies close to the object and its obstacles. *The length of the plan is unknown*. Each grasp turns the cap by at most the usable wrist range. An obstacle that blocks part of that range changes the number of grasps, i.e., the discrete structure of the solution, not only its geometry. *The task has memory*. The cap only moves while it is grasped and keeps its angle after release, so whether a goal can be reached depends on everything the robot has done before.

Existing planners each lack part of what this task needs. Siméon et al. [1] posed manipulation planning in the composite configuration space of robot and object. A plan alternates *transit*, where the robot moves alone, and *transfer*, where it moves the object at a fixed grasp, and the two can only meet where the grasp and placement spaces intersect. They built the resulting manipulation graph with probabilistic roadmaps. Sampling-based planners of this kind [1]–[3] are probabilistically complete: if a plan exists, the probability of finding it tends to one as the number of samples grows. After a finite time, however, they cannot conclude that no plan exists, or that the plan they return uses the fewest grasps. Since the sets where grasps happen have zero volume, they also need dedicated samplers to reach them at all. Optimisation-based task and motion planners [4], [5] fix a sequence of grasps and then solve a nonconvex program for the motion. The solver is local, so a failure may come from the initial guess rather than from infeasibility. To find the fewest grasps, sequences of increasing length are tried, but a failed shorter sequence is not a proof that it is infeasible. [CHECK: characterisation of regrasp maps [5].] Mixed-integer programs over convex decompositions of free space, used for footstep planning [6] and in our earlier manipulation planner [7], are complete and globally optimal relative to the decomposition. But they are solved for a fixed number of grasps, found by an outer loop, and their big-M constraints became numerically unreliable on a 7-DOF arm (Section IV-G). Learned policies take a different route. They twist lids off with two multi-fingered hands [8] or reorient objects in hand [9], without planning regrasps explicitly. They come with no completeness guarantee and cannot certify that a goal is unreachable. A policy trained on goals that need k regrasps has no guarantee on goals that need more, where errors compound over the longer horizon [10].

This work was supported by the UKRI AI Centre for Doctoral Training in Dependable and Deployable AI for Robotics (CDT-D2AIR) [TODO: grant number].

Footstep planning has the same combinatorial core as re-grasping: a sequence of contacts of unknown length, where each contact constrains the next. CASSR [11] showed that A* can search over convex sets of footholds reachable from the start, rather than over discretised footholds or integer variables, and plans 30 footsteps in under 125 ms. Regrasping needs more than a change of domain. Footholds lie on stepping stones that the environment provides as polygons. Grasping configurations, in contrast, must be built from nonlinear kinematics in a 10-dimensional space. A footstep does not move the terrain, whereas a grasp moves the object, so reachability must carry the state of the object. And the motions between grasps must avoid collisions in the full configuration space of the robot.

We represent the manipulation graph of [1] with convex sets. A convex decomposition of free space [12], [13] gives the *free sets*. A convex inner approximation of the grasping configurations, built from a seed grasp along the null space of the grasp constraints [14], gives the *grasp set*. Their intersections are the *transition sets*, in which the robot can grasp, turn and release the object. We search them with an A* whose nodes pair a convex set with the interval of object configurations reachable inside it. Each guarantee follows from one property of this representation:

- Every set is convex, so a robot that reaches a set can reach all of it along straight lines. No sampling is needed inside a set.
- The object moves only inside transition sets. When one object coordinate is coupled to one robot joint, as for caps, valves and knobs, the object configurations reachable in a set form an interval, which the search propagates and compares exactly.
- There are finitely many sets and the cost counts grasps. The search therefore terminates and returns a plan with the fewest grasps relative to the sets, or reports that none exists, without fixing the number of grasps in advance.
- Building the graph only requires linear programs, with no integer variables.

Our contributions are:

- *Transition sets*: a representation of the manipulation graph in which transit, transfer and the switches between them are convex sets. We build it for a 7-DOF arm from a convex decomposition and a null-space grasp set, and validate it against a collision checker where the decomposition is least reliable (Sections III and IV-A–IV-C).
- A *reachability* A* over transition sets. It is complete and returns the minimum number of grasps relative to the sets, with exact propagation and pruning when one object coordinate is coupled to one robot joint. A second stage turns its output into motions without a solver (Sections IV-D–IV-F).
- An evaluation on cap-turning tasks against our earlier mixed-integer planner and a sampling-based manipulation planner, up to the 10-dimensional space of a Franka arm,

hand and cap (Sections V–VI). **[TODO: update with E1–E5.]**

All guarantees hold relative to the free and grasp sets. They are only as meaningful as these sets are faithful near contact, which is why we validate the transition sets. Exact intervals rely on the coupling assumption of Section III. General objects are left to future work. **[CHECK: anonymise the self-citation if double-blind.]**

II. RELATED WORK

A. Regrasp and manipulation planning

The first regrasp planners searched over finite sets of grasps and placements [15], [16]. This continues in regrasp graphs over thousands of precomputed grasps [17], and in a bimanual planner that is certified complete relative to its discretisation [18]. Such planners are complete only with respect to the grasps they enumerate: a solution that needs a grasp between two discretised ones is missed. The manipulation graph [1] made grasps and placements continuous. It was extended to multi-modal planning [2], to asymptotically optimal sampling [19], [20], and to constraint graphs explored with constrained samplers [3], [21], [22]. All of these offer probabilistic or asymptotic guarantees, which hold in the limit of infinitely many samples. Task and motion planners add symbolic decisions on top [23], [24]. In optimisation-based TAMP, an outer search enumerates skeletons with a fixed number of mode switches, and a local solver handles each one [4]. Closest in spirit, Liu et al. [25] plan contact-rich planar manipulation on a graph of reachable sets of object poses **[CHECK: characterisation]**. Our transition sets are the nodes of the manipulation graph of [1], built as convex sets rather than sampled. This is what lets us guarantee completeness and the minimum number of grasps after a finite computation, relative to the sets.

B. Contact planning over convex sets

Footstep planning faces the same combinatorics of contact sequences of unknown length. A* over discretised footholds [26], [27] is complete only at the resolution of the discretisation. Mixed-integer programs over convex terrain [6], [28] avoid discretisation and are globally optimal, but the number of steps, or a bound on it, must be fixed, and their solve times are hard to bound. ℓ_1 relaxations [29], [30] make them fast at the cost of these guarantees. **[CHECK: SLIM characterisation.]** CASSR [11] instead searches with A* over convex sets of footholds reachable from the start, expanded by Minkowski sums, and places the footsteps with a quadratic program once the sets are chosen. CASSR discretises the yaw of the feet, and inflates its heuristic for speed, so its optimality is empirical rather than guaranteed. Our search follows the same principle in a different space. The reachable set of a node carries the configuration of the object rather than a foothold. In the coupled tasks we consider it is an interval, so we can keep an admissible heuristic and exact pruning without discretisation.

Graphs of Convex Sets (GCS) [31], [32] compute shortest paths over convex decompositions through a convex relaxation, and are globally optimal relative to the decomposition when the relaxation is tight. They plan the motion of a robot in one mode, with no switches of grasp and no object state carried from one switch to the next. They have been extended to contact-rich planar pushing [33]. Heuristic searches handle graphs too large to solve at once [34], [35]. GCS* is complete and optimal when it can decide exactly whether one partial path dominates another, i.e., whether its reachable set contains the other's at lower cost. In general this check must be approximated [34]. With interval reachable sets, it is exact.

C. Convex decompositions of free space

Convex decompositions inner-approximate free space with convex sets. IRIS grows them by alternating separating hyperplanes and inscribed ellipsoids [36]. C-IRIS certifies regions as collision-free with sums-of-squares programs [37], [38]. IRIS-NP, IRIS-NP2 and IRIS-ZO scale to more dimensions, with a probabilistic guarantee: the fraction of a region's volume in collision is below a threshold, with a given confidence [13], [39]. Clique covers automate the seeding [12], and GPU implementations approach online rates [40]. A volume guarantee says nothing about sets of near-zero volume, such as grasp sets. This is why we validate the transition sets separately (Section IV-C).

III. PROBLEM FORMULATION

A. Transit, transfer and transition sets

A robot with configuration $x \in \mathcal{X} \subset \mathbb{R}^{n_r}$ (arm and fingers) manipulates an object with configuration α in a known static scene. The state $q = (x, \alpha)$ lives in the composite configuration space $\mathcal{C}$ [1], and $\mathcal{C}_{\text{free}} \subseteq \mathcal{C}$ is the set of states without collisions other than the contacts a grasp requires. The object cannot move on its own. In the *placement space* $\mathcal{C}_P$ it rests stably, and in the *grasp space* $\mathcal{C}_G$ the robot holds it. A manipulation path alternates *transit* segments in $\mathcal{C}_P$, where the robot moves alone and α is constant, and *transfer* segments in $\mathcal{C}_G$, where the robot moves the object at a fixed relative pose. The two can only meet in the *transition sets* $\mathcal{C}_G \cap \mathcal{C}_P \cap \mathcal{C}_{\text{free}}$, where the robot can grasp or release the object. These are the nodes of the manipulation graph [1].

B. Multi-regrasp tasks with a coupled object coordinate

We study the tasks captured by the following assumption, which include caps, knobs, valves and cranks.

Assumption 1: (i) The object has one coordinate $\alpha \in [\alpha_{\min}, \alpha_{\max}]$ and is always stably placed, so $\mathcal{C}_P = \mathcal{C}$. (ii) Holding the object couples α to one robot joint w , the *wrist*: during a transfer, $\dot{\alpha} = -\dot{w}$. The sign follows from the orientation of the axes. (iii) Neither the grasp constraints nor collisions depend on α : $\mathcal{C}_G = G \times [\alpha_{\min}, \alpha_{\max}]$ and $\mathcal{C}_{\text{free}} = \mathcal{X}_{\text{free}} \times [\alpha_{\min}, \alpha_{\max}]$ for some $G, \mathcal{X}_{\text{free}} \subseteq \mathcal{X}$.

Point (iii) holds for rotationally symmetric objects, such as the cap in our experiments. By (i), every grasping configuration is also a placement, so the transition sets reduce to $G \cap \mathcal{X}_{\text{free}}$

(times the object range). Transfer motions also happen inside them.

The wrist has a limited range $[w_{\min}, w_{\max}]$, so turning the object by more than this range requires several grasps. Without loss of generality, let $\alpha_{\text{goal}} \leq \alpha_{\text{init}}$. Since α affects neither collisions nor grasps, a plan never gains from turning the object backwards, and the task works like a ratchet. A *stroke* grasps the object and moves the wrist forward, so α decreases by the wrist travel. A *return* releases the object and rotates the wrist back with α fixed, ready for the next stroke. A *release* lets go of the object before the robot leaves the transition set. Every other motion is transit in $\mathcal{X}_{\text{free}}$ with α fixed.

C. Problem statement

Given $q_{\text{init}} = (x_{\text{init}}, \alpha_{\text{init}})$ and $q_{\text{goal}} = (x_{\text{goal}}, \alpha_{\text{goal}})$, find a continuous path from q_{init} to q_{goal} in $\mathcal{C}_{\text{free}}$, made of transit segments and strokes, that uses the minimum number of grasps. Ties are broken by the number of changes of convex set along the path. Path length is not optimised.

In terms of reachability, let $\mathcal{R}_k \subseteq \mathcal{C}_{\text{free}}$ be the set of states reachable from q_{init} with at most k grasps. The minimum number of grasps is the smallest k such that $q_{\text{goal}} \in \mathcal{R}_k$, and the goal is unreachable if there is no such k . Neither $\mathcal{R}_k$ nor $\mathcal{C}_{\text{free}}$ can be computed exactly. Our planner computes $\mathcal{R}_k$ exactly for an inner approximation of $\mathcal{X}_{\text{free}}$ and G by convex polytopes, and its guarantees hold relative to that approximation.

IV. METHOD

The planner has an offline stage, which depends only on the robot, the object and the scene, and an online stage that is run for each query. Offline, it builds the free sets (Section IV-A) and the grasp set (Section IV-B), and intersects them into validated transition sets and a graph, using only linear programs (Section IV-C). Online, the reachability A* chooses the sequence of sets and grasps (Section IV-D), and a second stage turns it into motions (Section IV-E). Fig. 1 illustrates the online stage. New goals and start configurations reuse the offline stage unchanged.

A. Free sets

We inner-approximate $\mathcal{C}_{\text{free}}$ by polytopes $P_j = \{q : A_j q \leq b_j\}$, $j = 1, \dots, J$, computed with clique covers of visibility graphs [12] and IRIS-NP2 regions [13] in Drake [41]. By Assumption 1(iii), the object coordinate does not affect collisions. We therefore replace each region by its slice at $\alpha = 0$, $F_j = \{x : (x, 0) \in P_j\}$. This is sound: if $(x, 0)$ is collision-free, so is (x, α) for every α . The robot can then occupy any $x \in F_j$ at any object angle, and the planner tracks α separately.

B. Grasp set

For a 7-DOF arm, a grasp constrains the pose of the hand frame E relative to the cap frame C , a nonlinear function of the arm joints. The hand must be centred above the cap,

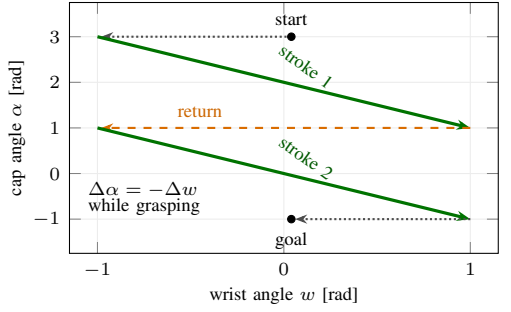
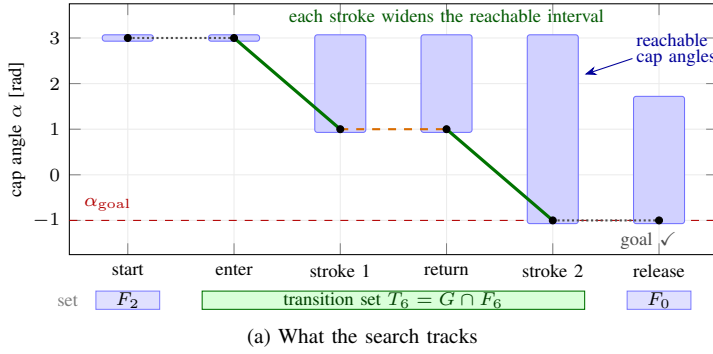


Fig. 1. Reachability A^* on the 10-DOF cap task (query Q0): turn the cap from $\alpha = 3$ to -1 rad and bring the arm back to its start. (a) Each column is one node on the returned path: the set the robot is in (strip) and the interval of cap angles reachable there (bar). The black line is the cap angle along the final plan. Entering the transition set T_6 changes nothing; each stroke widens the interval by up to the wrist range of T_6 (2 rad); returns and transit keep it. Leaving T_6 towards F_0 needs some further wrist motion, which lowers the top of the last bar. The search stops at the first node whose set contains the goal arm configuration and whose interval contains α_{goal} . (b) While grasping, the cap turns as the wrist moves forward; returns move the wrist back with the cap fixed. Two strokes are the minimum, since each turns the cap by at most 2 rad. **[TODO: regenerate once the transition sets are validated (E5).]**

within a height band, with its approach axis $z_E = {}^C R^E(x) e_z$ pointing down:

$$h(x) = [p_x, p_y, z_{E,x}, z_{E,y}]^T = 0, \quad (1)$$

$$g(x) = [p_z - z_{\max}, z_{\min} - p_z, z_{E,z} + c]^T \leq 0, \quad (2)$$

with $p = {}^C p^E(x)$, $[z_{\min}, z_{\max}] = [0.105, 0.110]$ m and $c = 0.9$. The fingers are restricted to a 1 mm band around the width of the cap. The set $\{h = 0, g \leq 0\}$ is thin and curved, so it cannot be written as a polytope. We approximate it locally, using two ideas from NLP sampling [14]. First, a seed grasp x_0 is found by Gauss–Newton minimisation of the constraint violation $\| [h(x); \max(g(x), 0)] \|^2$ from a random start. Second, the directions that preserve the grasp to first order are the null space of the Jacobian J_h of the equality constraints, which NLP sampling uses to move along constraint manifolds. The grasp set keeps the configurations that stay within ϵ of this null space, around x_0 :

$$G = \{x : |h_0 + J_h(x - x_0)| \leq \epsilon, \quad g_0 + J_g(x - x_0) \leq \epsilon, \\ x_{\min}^G \leq x \leq x_{\max}^G, \quad |x_j - x_{0,j}| \leq \rho \quad \forall j \in \mathcal{A}\}, \quad (3)$$

where h_0, J_h, g_0, J_g are evaluated at x_0 , $\epsilon = 10^{-3}$, and $\mathcal{A}$ are the arm joints other than the wrist.

Two properties of the task shape this set. First, the wrist axis passes through the origin of E along z_E , so rotating the wrist changes neither p nor z_E . The grasp constraints are therefore *exactly* invariant along w , which we verified to machine precision over the whole wrist range, and G spans the full wrist range. Second, J_h has rank 4, so the null space has 6 dimensions, and without the trust region ρ in (3) it would only be bounded by the joint limits. Far from x_0 , the linearisation is meaningless: without the trust region, the extreme points used by the graph moved an arm joint by 2.27 rad from x_0 , with $\|h\|_\infty = 0.95$ and the hand 0.80 m away from the cap. With $\rho = 0.05$ rad, $\|h\|_\infty$ stays around 2×10^{-3} along our plans **[CHECK: numbers from the guide's q0 run]**. Seeds on other parts of the grasp manifold give further grasp sets that

the search treats in the same way. We use a single seed in this paper (see Section VII).

C. Transition sets and the graph

For every free set with $F_k \cap G \neq \emptyset$, we form the transition set $T_k = G \cap F_k$. Two LPs give its extreme wrist values $w_k^\pm = \max / \min \{\pm w : x \in T_k\}$ and points $x_k^\pm \in T_k$ attaining them. A stroke inside T_k turns the object by at most $L_k = w_k^+ - w_k^-$, and we discard sets with $L_k < L_{\min}$. Since G is thin in every direction but the wrist, a transition set is essentially a wrist interval, and by convexity the segment $[x_k^-, x_k^+]$ lies in T_k .

a) *Validation*: The free sets are collision-free only in a volume sense: IRIS-NP2 bounds the fraction of each region's volume in collision [13]. A transition set has almost no volume, so it can be entirely in collision without violating that bound, and it lies exactly where the hand is closest to the object and the obstacles. We therefore validate each T_k separately. Since T_k is essentially a segment, we check densely spaced configurations on $[x_k^-, x_k^+]$ with a collision checker, and replace T_k by the sub-intervals of the wrist on which every check passes. Each of these is T_k intersected with a bound on w , and is therefore still a polytope. **[TODO: implement (E5); state the check resolution; this is a dense check, not a certificate. Certifying these thin sets with C-IRIS [38] is an alternative.]**

b) *Graph*: The graph has two kinds of edges. A *transit* edge joins F_i and F_j if they intersect, and its door point is the Chebyshev centre of $F_i \cap F_j$. A *switch* edge joins F_i and T_k if they intersect. For it, we store the wrist range $[a_{ik}^-, a_{ik}^+]$ of $F_i \cap T_k$ and the points attaining its ends. Each of these quantities is an LP feasibility check, a Chebyshev centre or an extreme point. No integer variables or vertex enumeration are needed, and the number of LPs is $O(J^2 + JK)$ for J free sets and K transition sets.

D. Reachability A^*

a) *Nodes*: As in CASSR [11], a node does not represent a configuration but a set reachable from the start. A *free*

TABLE I
MOVES OF THE REACHABILITY A*. $c_1 = w_k^+ - a^-$ AFTER ENTERING,
 $c_1 = L_k$ AFTER A RETURN; $d = \max(0, a_{jk}^- - a^+)$.

Move	From $\rightarrow$ to	Interval update	Cost
transit	$F_i \rightarrow F_j, F_i \cap F_j \neq \emptyset$	–	μ
enter	$F_i \rightarrow T_k$ (entered), $F_i \cap T_k \neq \emptyset$	–	μ
stroke	entered/released $\rightarrow$ holding	$\alpha^- \leftarrow \alpha^- - c_1$	1
return	holding $\rightarrow$ released	–	1
release	holding $\rightarrow F_j, F_j \cap T_k \neq \emptyset$	$\alpha^+ \leftarrow \alpha^+ - d$	1
leave	entered $\rightarrow F_j$, only if $d = 0$	–	μ

node $(F_i, [\alpha^-, \alpha^+])$ states that every $x \in F_i$ is reachable together with every $\alpha \in [\alpha^-, \alpha^+]$. Since F_i is convex and α is constant during transit, any point of F_i can be reached from any other along a straight line. A *transition node* $(T_k, [\alpha^-, \alpha^+], \phi, [a^-, a^+])$ also records the wrist range $[a^-, a^+]$ through which the set was entered, and a phase ϕ : *entered* (in the set, not yet grasping), *holding* (just completed a stroke) or *released* (just completed a return, with the wrist back at w_k^-). The interval carries all the information about the path so far. For example, reaching F_3 before any grasp and after a stroke of 2rad gives the same set with the intervals $\{3\}$ and $[1, 3]$, which are two different nodes.

b) *Moves and costs*: Table I lists the successors of a node. Only strokes widen the interval. A stroke can turn the object by any amount up to its maximum, so it lowers α^- and keeps α^+ . Since the object can only be grasped inside transition sets, these are the only sets where reachability grows in α . Leaving a transition set towards F_j can also force some turning. Inside T_k , the wrist only moves forward while grasping, so reaching the exit range from the entry range turns the object by at least $d = \max(0, a_{jk}^- - a^+)$, which lowers α^+ . We clip α^- at $\max(\alpha_{\text{goal}}, \alpha_{\text{min}})$. Strokes, returns and releases cost 1 each, and every other move costs $\mu \ll 1$ ($\mu = 10^{-3}$). Every stroke is followed by exactly one return or release, so the cost of a plan is twice its number of grasps, plus μ times its number of set changes. Minimising the cost therefore minimises the number of grasps first and the set changes second. Releasing must cost 1, otherwise leaving and re-entering a transition set would be a cheaper substitute for a return. **[CHECK: the code also allows a return from the entered phase at cost 1, before any grasp; such a move should cost μ for the cost to be exactly twice the number of grasps.]**

c) *Heuristic, goal and dominance*: With $L_{\text{max}} = \max_k L_k$, the heuristic

$$h(\nu) = 2 \left\lceil \frac{\alpha^- - \alpha_{\text{goal}}}{L_{\text{max}}} \right\rceil + \mathbf{1}[\phi = \text{holding}] \quad (4)$$

is admissible: each remaining stroke turns the object by at most L_{max} and is followed by a return or the final release, and a holding node must still release. It is also consistent, since no move lowers h by more than its cost. A free node is a goal if $x_{\text{goal}} \in F_i$ and $\alpha_{\text{goal}} \in [\alpha^-, \alpha^+]$. A node is *dominated*, and skipped, if an expanded node with the same key has an interval that contains its interval at no higher cost. The key

Algorithm 1 Reachability A*

Input: graph of Sec. IV-C; $(x_{\text{init}}, \alpha_{\text{init}})$, $(x_{\text{goal}}, \alpha_{\text{goal}})$

Output: sequence of sets and grasp actions, or NONE

```

1:  $\mathcal{O} \leftarrow \{(F_i, [\alpha_{\text{init}}, \alpha_{\text{init}}]) : x_{\text{init}} \in F_i\}$ ;  $\mathcal{E} \leftarrow \emptyset$ 
2: while  $\mathcal{O} \neq \emptyset$  do
3:    $\nu \leftarrow \arg \min_{\nu \in \mathcal{O}} g(\nu) + h(\nu)$ ; remove  $\nu$  from  $\mathcal{O}$ 
4:   if  $\nu$  dominated by  $\mathcal{E}$  then continue
5:    $\mathcal{E} \leftarrow \mathcal{E} \cup \{\nu\}$ 
6:   if  $\nu = (F_i, [\alpha^-, \alpha^+])$ ,  $x_{\text{goal}} \in F_i$ ,  $\alpha_{\text{goal}} \in [\alpha^-, \alpha^+]$ 
     then
7:     return path to  $\nu$ 
8:   for each successor  $\nu'$  of  $\nu$  (Table I) do
9:     if  $\nu'$  not dominated by  $\mathcal{E}$  then add  $\nu'$  to  $\mathcal{O}$ 
10: return NONE ▷ no plan within the sets

```

is the set for free nodes, and the set, phase and entry range for transition nodes. Because the key and the interval describe everything that can still be reached from a node, this check is exact. Algorithm 1 summarises the search.

E. From sets to motions

The search fixes the sets and the grasp actions, for example *enter*, *stroke*, *return*, *stroke*, *release*. As in CASSR, a second stage computes the configurations, here without a solver. A backward pass from α_{goal} first decides how much each visit to a transition set turns the object, within the maximum turn of each of its strokes. It ends exactly at α_{init} , because the search proved α_{goal} reachable. Waypoints are then placed at the door points for transit, at points of $F_i \cap T_k$ for entering and leaving, and on the segment $[x_k^-, x_k^+]$ for strokes and returns. Each motion between two waypoints is a straight line, and we record a set, F_j or T_k , that contains both of its ends. By convexity, that set contains the whole motion. We check this containment and the coupling ($\Delta\alpha = -\Delta w$ on strokes, $\Delta\alpha = 0$ otherwise) for every motion. Path length is not optimised. Since the sequence of sets is fixed, a convex program over these sets, e.g., GCS [32] restricted to them, could shorten the path. **[TODO: implement, or keep as future work.]**

F. Properties

Proposition 1 (informal): Relative to the free sets $\{F_j\}$ and the transition sets $\{T_k\}$: (i) every motion returned is a straight line inside one set that satisfies the coupling of Assumption 1, so the path is collision-free whenever the sets are; (ii) the search terminates. If a plan exists whose transit motions lie in the free sets and whose strokes lie in the transition sets, it returns one with the minimum number of grasps, with ties broken by the number of set changes. Otherwise, it returns NONE; (iii) the second stage succeeds on every sequence the search returns.

Sketch. (i) follows from convexity of the recorded sets. (ii) Each stroke lowers α^- by at least L_{min} until it reaches α_{goal} , so finitely many intervals arise per key, and the search terminates. With a consistent heuristic and exact dominance,

A* expands a goal node of minimum cost first [42]. Entering at the highest wrist value and leaving at the lowest wrist value of each switch edge minimises forced turning, so restricting entries and exits to these points loses no plan. (iii) The backward pass mirrors the interval updates of Table I. **[CHECK: write the full proof; (iii) is asserted in our implementation but not yet proven. Overlapping transition sets would need chain-aware placement, so each set is its own segment for now.]**

□

These guarantees are relative: a plan reported as optimal uses the fewest grasps *that the sets allow*. This is why the transition sets must be validated (Section VI-C).

G. Relation to a mixed-integer formulation

Our earlier planner [7] posed the same problem as an MIQP. For a fixed number of grasps ℓ , it chose keyframes $q_{\text{init}}, x_1, \dots, x_{2\ell}, q_{\text{goal}}$, with binary variables for the set containing each keyframe and big-M constraints $A_j x_i \leq b_j + M(1 - z_{ij})$, and solved for $\ell = 1, 2, \dots$ until feasible. This works on 3- and 4-DOF tasks. On the 10-DOF task, the grasp set is $2\epsilon = 2 \times 10^{-3}$ thick, whereas with $M = 10^6$ and MOSEK's default integrality tolerance of 10^{-5} [43], a binary accepted as 1 may relax its constraints by $M \cdot 10^{-5} = 10$ rad, more than any joint range. Tight big-M constants or convex-hull formulations [44] would mitigate this at the cost of a larger model, but the number of grasps would still be fixed by an outer loop. **[CHECK: record the MOSEK status/log for the 10-DOF MIQP failure.]**

V. EXPERIMENTAL EVALUATION

Our experiments ask four questions. Does the search return plans with the minimum number of grasps for new goals, start configurations and obstacle layouts, from a single offline stage? How does it compare with our earlier MIQP and with a sampling-based manipulation planner as the number of required grasps grows? Why must the transition sets be validated? And does validation change the plans?

A. Tasks and implementation

Both tasks turn a cap about a fixed vertical axis with a Franka Panda [45], whose wrist is restricted to $[-1, 1]$ rad, so any turn larger than 2 rad needs several grasps.

- **T1 (3-DOF)**. All arm joints but the wrist are locked: $q = (w, x_{\text{finger}}, \alpha)$, with $x_{\text{finger}} \in [-0.045, 0]$ m and $\alpha \in [-\pi, \pi]$. The decomposition has 15 regions, and the task turns the cap by 2π . **[CHECK: T1 from the MInf report (WSG gripper) or the locked Franka?]**
- **T2 (10-DOF)**. Seven arm joints, two fingers and the cap. The cap is 0.5 m in front of the robot base, with two cylindrical obstacles (radius 5 mm) next to it. The free sets come from an IRIS-NP2 clique cover with 42 regions **[CHECK: coverage target 0.98; (ϵ, δ) ; generation time]**. The grasp set uses (3) with $\epsilon = 10^{-3}$, $\rho = 0.05$ rad and $L_{\min} = 0.05$ rad. The reference query Q0 turns the cap from 3 to -1 rad and returns the arm to its start configuration.

The planner is implemented in Python with Drake [41] (v1.48). The LPs use Drake's `MathematicalProgram`, and the MIQP uses MOSEK [43]. **[TODO: hardware; code release.]**

B. Baselines and protocol

We compare against our earlier MIQP [7] (Section IV-G), with GCS [32] connecting its keyframes, and against a sampling-based planner on the manipulation graph [1], with probabilistic roadmaps for transit and transfer. **[TODO: implement the sampling-based baseline (own implementation or HPP [3]); time limit.]** For all methods, every returned motion is checked densely with Drake's collision checker. **[TODO: Run E1–E5; this is the evidence for the main claim.]** **E1, goal sweep:** α_{goal} on a grid over the cap range, with the arm configuration fixed. We report the number of grasps against the lower bound $\lceil D/L_{\max} \rceil$ for a turn D , and the search time. **E2, random queries:** N start and goal arm configurations sampled in the free sets, with random turning angles. **E3, obstacle layouts:** M placements of the obstacles, each with its own offline stage. We report offline time (decomposition, grasp set, transition sets, graph) and online time per query. **E4, baselines:** all three planners as a function of the number of grasps the query requires, on T1 and T2. We report success within the time limit, planning time and number of grasps. **E5, validation:** Q0 with and without the validation of Section IV-C.

VI. RESULTS

A. Reachability A* on the 10-DOF task

Table II and Fig. 1 report Q0, before validation. Of the 42 free sets, 8 intersect the grasp set. Four of these transition sets span the whole wrist range ($L_k = 2$ rad), and the others span between 0.06 and 0.36 rad. The search expands 63 nodes in 4 ms and returns two strokes, one return and one release: it enters T_6 from F_2 , performs stroke, return and stroke, and releases into F_0 . Its cost, $4 + \mu$, equals the initial heuristic plus one set change, so the heuristic is tight on this query. With $D = 4$ rad and $L_{\max} = 2$ rad, two grasps is the minimum that the sets allow. The second stage turns 2 rad per stroke and produces 7 waypoints, and all 6 straight-line motions lie in their recorded set and satisfy the coupling. The online stage never calls a collision checker: its only geometric operations are the LPs of the offline stage.

B. Comparison with the MIQP

Table III compares the two ways of choosing grasps. On T1, the MIQP returns 11 grasps after proving 10 infeasible. Each stroke turns the cap by 0.604 rad, the wrist range of the grasp set in that decomposition, and $\lceil 2\pi/0.604 \rceil = 11$. GCS then connects all 23 transit and transfer segments. On T2, the MIQP does not return a reliable solution, for the reasons of Section IV-G, whereas the search returns the plan above in milliseconds. **[CHECK: MIQP outcome and solve time on T2; run the reachability A* on T1 and check that it also returns 11.]**

TABLE II
REACHABILITY A* ON T2 (10-DOF), QUERY Q0, BEFORE VALIDATION

Stage	Quantity	Value
Offline	free sets J (non-empty after slicing)	42
	transition sets K (with $L_k = 2$ rad)	8 (4)
	transit / switch edges	850 / 58
	graph construction time	[TBD]
	decomposition time	[TBD]
Online	nodes expanded	63
	search time	4 ms
	strokes / returns / releases	2 / 1 / 1
	set changes	1
	motions contained in their set	6 / 6

TABLE III
CHOOSING GRASPS WITH THE MIQP AND WITH THE REACHABILITY A*

Task	Method	Grasps	Time	Outcome
T1 (3-DOF)	MIQP	11	[TBD]	solved; GCS connected
	Reach. A*	[TBD]	[TBD]	[TBD]
T2 (10-DOF)	MIQP	–	[TBD]	numerical failure [CHECK:]
	Reach. A*	2	4 ms	solved (rel. sets)

C. Why transition sets must be validated

A planner that is complete and optimal relative to its sets inherits their errors. To measure them where they matter, we swept the wrist over $[-1, 1]$ at the seed grasp, at 99 points, and compared membership in the free sets with Drake’s collision checker. The checker finds the wrist free only on $[-1, -0.19]$ and $[0.70, 0.92]$ rad [CHECK: which bodies collide in between: fingers vs. obstacles?]. All four decompositions we computed for T2 (IRIS-NP2 with 42 and 46 regions, and two IRIS-ZO covers with 64 regions [CHECK: the two IRIS-ZO files may be the same cover]) contain every point of the sweep: 53 true positives, 46 false positives and no false negatives in each case. The two-grasp plan of Fig. 1 is therefore valid relative to the sets, but collides in reality. With the longest truly free wrist interval of 0.81 rad, the minimum is $\lceil 4/0.81 \rceil = 5$ grasps: a different discrete structure, not just a different geometry.

This is not a bug in the region generators, but a mismatch between their guarantee and our use of them. IRIS-NP2 and IRIS-ZO bound the *fraction of volume* of a region in collision [13]. A transition set has near-zero volume, so it can be entirely in collision without violating that bound, and grasps happen exactly where the hand is closest to the obstacles. The same sensitivity appeared on low-dimensional versions of the task [7]: with a 4-DOF gripper, a decomposition at 0.70 coverage contained no grasp–release sequence, and at 0.90 it contained one that GCS could not connect. For regrasp planning, what matters is not coverage of free space but the fidelity of the transition sets, which is what the validation step checks. [TODO: E5: Q0 with validated transition sets; expected 5 grasps with no change to the search.]

D. Generalisation across goals, starts and scenes

(left) E1: grasps vs. turning angle D , a staircase tracking $\lceil D/L \rceil$; (right) E4: planning time vs. required grasps for the reachability A*, the MIQP and the sampling-based baseline

Fig. 2. [TODO: E1 and E4 results.]

[TODO: Results of E1–E4 (Section V-B).]

VII. DISCUSSION AND LIMITATIONS

a) *Validating where it matters*: The weakest link is not the search but the sets it searches over (Section VI-C). Transition sets are thin and almost one-dimensional, so they are much cheaper to validate than full regions, and the search only needs their wrist intervals, so validation does not change it. A dense check is not a certificate. Certifying transition sets with C-IRIS [38], or seeding the decomposition on the grasp set so that it is dense where grasps happen, would strengthen the guarantees.

b) *Constructing grasp sets*: The grasp set uses only part of NLP sampling [14]: its Gauss–Newton phase to find a seed, and its null-space view of the constraints. We also tried to use its interior samplers, nonlinear hit-and-run and manifold RRT, to explore the grasp manifold and build grasp sets from the samples, but neither gave usable polytopes. Growing IRIS regions around grasp samples produced a very large number of very small polytopes, since a region grown around a thin, curved set is limited by its curvature in every direction. Fitting a union of polytopes to the samples with a small network, in which each neuron is a half-space, a region is the maximum over its half-spaces and the set is the minimum over regions, did not represent the grasp set accurately enough to plan with. On the grasp problem, hit-and-run also mixed poorly in the wrist and cap coordinates, because its line search shares one step size across all coordinates. The construction we use instead exploits the structure of the constraints, in particular their exact invariance to the wrist, rather than discovering it from samples. Its limitation is that it is local: one seed gives one grasp set, valid near the seed only. The restarts of NLP sampling, which spread seeds over the components of the grasp manifold, are a natural way to obtain several grasp sets, which the search already supports. [TODO: numbers for the attempts, if space allows.]

c) *Scope*: Exact intervals rely on Assumption 1, which covers caps, knobs, valves and cranks but not general pick-and-place. With more object coordinates, the object configurations reachable in a set become sets that could be propagated with Minkowski sums and intersections, as in CASSR [11], at the cost of exact dominance and with nonlinear grasp transforms to approximate.

d) *Optimality and offline cost:* The search minimises the number of grasps, and then the number of set changes, a weak proxy for path length. As in CASSR, the sets are fixed before the configurations, so path length is not optimal. The heuristic uses the longest transition set in the graph; one computed per connected component would be tighter. A new obstacle layout requires a new decomposition, which dominates the offline cost: in a separate run, IRIS-NP2 took 120s for 25 regions at a 0.98 coverage target [CHECK: different settings]. GPU implementations [40] reduce this cost, and new start and goal queries reuse the graph unchanged.

VIII. CONCLUSION

Tasks such as turning a cap, a valve or a knob require a robot to grasp the same object many times, and the number of grasps depends on the goal, the kinematics and the obstacles. We represented the manipulation graph with convex sets, so that the switches between moving alone and moving the object are convex transition sets, and searched them with an A* whose nodes pair a set with the interval of object configurations reachable inside it. For objects with one coordinate coupled to the wrist, this search is complete and returns the minimum number of grasps relative to the sets, without fixing that number in advance and without integer variables. It brings to regrasping the idea, matured in footstep planning, of searching over convex reachable sets rather than individual contacts, extended to contact sets that must be constructed and to contacts that move the object. Its guarantees are only as good as the transition sets near contact, which is why we validate them. [TODO: one sentence on E1–E5 once run.]

ACKNOWLEDGMENT

[TODO: acknowledgments; funding is in the title footnote.]

REFERENCES

- [1] T. Siméon, J.-P. Laumond, J. Cortés, and A. Sahbani, “Manipulation planning with probabilistic roadmaps,” *The International Journal of Robotics Research*, vol. 23, no. 7–8, pp. 729–746, 2004.
- [2] K. Hauser and J.-C. Latombe, “Multi-modal motion planning in non-expansive spaces,” *The International Journal of Robotics Research*, vol. 29, no. 7, pp. 897–915, 2010.
- [3] J. Mirabel, S. Tonneau, P. Fernbach, A.-K. Seppälä, M. Campana, N. Mansard, and F. Lamiriaux, “HPP: A new software for constrained motion planning,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016, pp. 383–389.
- [4] M. Toussaint, “Logic-geometric programming: An optimization-based approach to combined task and motion planning,” in *Proc. International Joint Conference on Artificial Intelligence (IJCAI)*, 2015, pp. 1930–1936.
- [5] S. Levit and M. Toussaint, “Regrasp maps for sequential manipulation planning,” *arXiv preprint arXiv:2507.12407*, 2025.
- [6] R. Deits and R. Tedrake, “Footstep planning on uneven terrain with mixed-integer convex optimization,” in *Proc. IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2014, pp. 279–286.
- [7] J. López Gómez, “Optimisation-based manipulation planning in convex decompositions of C-free,” MInf Project Report, School of Informatics, University of Edinburgh, 2025.
- [8] T. Lin, Z.-H. Yin, H. Qi, P. Abbeel, and J. Malik, “Twisting lids off with two hands,” in *Proc. Conference on Robot Learning (CoRL)*, 2024.
- [9] OpenAI, M. Andrychowicz, B. Baker *et al.*, “Learning dexterous in-hand manipulation,” *The International Journal of Robotics Research*, vol. 39, no. 1, pp. 3–20, 2020.
- [10] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2011, pp. 627–635.
- [11] J. Wang and S. Tonneau, “CASSR: Continuous A-star search through reachability for real time footstep planning,” *arXiv preprint arXiv:2603.02989*, 2026.
- [12] P. Werner, A. Amice, T. Marcucci, D. Rus, and R. Tedrake, “Approximating robot configuration spaces with few convex sets using clique covers of visibility graphs,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [13] P. Werner, T. Cohn, R. H. Jiang, T. Seyde, M. Simchowitz, R. Tedrake, and D. Rus, “Faster algorithms for growing collision-free convex polytopes in robot configuration space,” in *International Symposium on Robotics Research (ISRR)*, 2024.
- [14] M. Toussaint, C. V. Braun, and J. Ortiz-Haro, “NLP sampling: Combining MCMC and NLP methods for diverse constrained sampling,” *arXiv preprint arXiv:2407.03035*, 2024.
- [15] P. Tournassoud, T. Lozano-Pérez, and E. Mazer, “Regrasping,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 1987, pp. 1924–1928.
- [16] R. Alami, T. Siméon, and J.-P. Laumond, “A geometrical approach to planning manipulation tasks. the case of discrete placements and grasps,” in *The Fifth International Symposium on Robotics Research*. MIT Press, 1990, pp. 453–463.
- [17] W. Wan and K. Harada, “Regrasp planning using 10,000s of grasps,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 1929–1936.
- [18] P. Lertkultanon and Q.-C. Pham, “A certified-complete bimanual manipulation planner,” *IEEE Transactions on Automation Science and Engineering*, vol. 15, no. 3, pp. 1355–1368, 2018.
- [19] P. S. Schmitt, W. Neubauer, W. Feiten, K. M. Wurm, G. v. Wichert, and W. Burgard, “Optimal, sampling-based manipulation planning,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2017.
- [20] W. Vega-Brown and N. Roy, “Asymptotically optimal planning under piecewise-analytic constraints,” in *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2016.
- [21] D. Berenson, S. Srinivasa, and J. Kuffner, “Task space regions: A framework for pose-constrained manipulation planning,” *The International Journal of Robotics Research*, vol. 30, no. 12, pp. 1435–1460, 2011.
- [22] Z. Kingston, M. Moll, and L. E. Kavraki, “Sampling-based methods for motion planning with constraints,” *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 1, pp. 159–185, 2018.
- [23] C. R. Garrett, R. Chitnis, R. Holladay, B. Kim, T. Silver, L. P. Kaelbling, and T. Lozano-Pérez, “Integrated task and motion planning,” *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 4, pp. 265–293, 2021.
- [24] C. R. Garrett, T. Lozano-Pérez, and L. P. Kaelbling, “PDDLStream: Integrating symbolic planners and blackbox samplers via optimistic adaptive planning,” in *Proc. International Conference on Automated Planning and Scheduling (ICAPS)*, vol. 30, 2020, pp. 440–448.
- [25] S. Liu, T. Zhao, B. P. Graesdal, P. Werner, J. Wang, J. Dolan, C. Liu, and T. Pang, “Approximately optimal global planning for contact-rich SE(2) manipulation on a graph of reachable sets,” *arXiv preprint arXiv:2601.10827*, 2026.
- [26] J. Chestnutt, M. Lau, G. Cheung, J. Kuffner, J. Hodgins, and T. Kanade, “Footstep planning for the Honda ASIMO humanoid,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2005, pp. 629–634.
- [27] R. J. Griffin, G. Wiedebach, S. McCrory, S. Bertrand, I. Lee, and J. Pratt, “Footstep planning for autonomous walking over rough terrain,” in *Proc. IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2019, pp. 9–16.
- [28] B. Aceituno-Cabezas, C. Mastalli, H. Dai *et al.*, “Simultaneous contact, gait, and motion planning for robust multilegged locomotion via mixed-integer convex optimization,” *IEEE Robotics and Automation Letters*, vol. 3, no. 3, pp. 2531–2538, 2018.
- [29] S. Tonneau, D. Song, P. Fernbach, N. Mansard, M. Taïx, and A. Del Prete, “SL1M: Sparse L1-norm minimization for contact planning on uneven terrain,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 6604–6610.
- [30] D. Song, P. Fernbach, T. Flayols, A. Del Prete, N. Mansard, S. Tonneau, and Y. J. Kim, “Solving footstep planning as a feasibility problem using

- L1-norm minimization,” *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 5961–5968, 2021.
- [31] T. Marcucci, J. Umenberger, P. Parrilo, and R. Tedrake, “Shortest paths in graphs of convex sets,” *SIAM Journal on Optimization*, vol. 34, no. 1, pp. 507–532, 2024.
 - [32] T. Marcucci, M. Petersen, D. von Wrangel, and R. Tedrake, “Motion planning around obstacles with convex optimization,” *Science Robotics*, vol. 8, no. 84, p. eadf7843, 2023.
 - [33] B. P. Graesdal, S. Y. C. Chia, T. Marcucci, S. Morozov, A. Amice, P. A. Parrilo, and R. Tedrake, “Towards tight convex relaxations for contact-rich manipulation,” in *Proc. Robotics: Science and Systems (RSS)*, 2024.
 - [34] S. Y. C. Chia, R. H. Jiang, B. P. Graesdal, L. P. Kaelbling, and R. Tedrake, “GCS*: Forward heuristic search on implicit graphs of convex sets,” in *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2024.
 - [35] R. Natarajan, C. Liu, H. Choset, and M. Likhachev, “Implicit graph search for planning on graphs of convex sets,” in *Proc. Robotics: Science and Systems (RSS)*, 2024.
 - [36] R. Deits and R. Tedrake, “Computing large convex regions of obstacle-free space through semidefinite programming,” in *Algorithmic Foundations of Robotics XI*. Springer, 2015, pp. 109–124.
 - [37] A. Amice, H. Dai, P. Werner, A. Zhang, and R. Tedrake, “Finding and optimizing certified, collision-free regions in configuration space for robot manipulators,” in *Algorithmic Foundations of Robotics XV (WAFR 2022)*. Springer, 2023, pp. 328–348.
 - [38] H. Dai, A. Amice, P. Werner, A. Zhang, and R. Tedrake, “Certified polyhedral decompositions of collision-free configuration space,” *The International Journal of Robotics Research*, vol. 43, no. 9, pp. 1322–1341, 2024.
 - [39] M. Petersen and R. Tedrake, “Growing convex collision-free regions in configuration space using nonlinear programming,” *arXiv preprint arXiv:2303.14737*, 2023.
 - [40] P. Werner, R. Cheng, T. Stewart, R. Tedrake, and D. Rus, “Superfast configuration-space convex set computation on GPUs for online motion planning,” *arXiv preprint arXiv:2504.10783*, 2025.
 - [41] R. Tedrake and the Drake Development Team, “Drake: Model-based design and verification for robotics,” <https://drake.mit.edu>, 2019.
 - [42] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.
 - [43] MOSEK ApS, “The MOSEK optimization toolbox,” <https://www.mosek.com>, 2024.
 - [44] J. P. Vielma, “Mixed integer linear programming formulation techniques,” *SIAM Review*, vol. 57, no. 1, pp. 3–57, 2015.
 - [45] S. Haddadin, S. Parusel, L. Johansmeier, S. Golz, S. Gabl, F. Walch, M. Sabaghian, C. Jähne, L. Hausperger, and S. Haddadin, “The Franka Emika robot: A reference platform for robotics research and education,” *IEEE Robotics & Automation Magazine*, vol. 29, no. 2, pp. 46–64, 2022.